

WayCool: System Architecture Document

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Abstract

This document presents the comprehensive system architecture for WayCool, India's leading full-stack food supply chain platform focused on B2B fresh produce distribution, farmer financing, cold chain infrastructure, and data-driven advisory. The architecture is designed to support 4 million+ farmers, 60,000+ B2B customers, 20,000+ tonnes of daily produce volume, 10,000+ IoT sensors, Rs.6,000 crore in farmer financing AUM, and Rs.12,000+ crore in annual revenue by FY30F.

The architecture follows a microservices-based, cloud-native design with offline-first capabilities for rural connectivity and IoT-first data ingestion from cold chain sensors. Key components include: (1) multi-channel farmer/B2B customer onboarding system, (2) WayCool Credit Engine for underwriting (78% default prediction accuracy), (3) NBFC lending and collections module, (4) cold chain monitoring platform with real-time temperature alerting, (5) Outgrow contract farming management system, (6) WayCool Lens farmer advisory service via WhatsApp/SMS, (7) B2B e-commerce and order management platform, (8) warehouse and logistics optimization engine, (9) data monetization platform for enterprise customers, and (10) analytics and reporting layer.

The architecture emphasizes scalability (horizontal scaling to 10M+ daily API requests), fault tolerance (99.9% uptime target, 4-hour RTO), security (AES-256 encryption, RBAC, mTLS for IoT devices), and compliance with RBI NBFC regulations, DPDP Act (India), and FSSAI quality standards.

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1 Architecture Overview

WayCool's system architecture is built on a microservices-based, cloud-native design deployed on AWS (Mumbai region) with edge nodes for offline operation in low-connectivity rural areas. The architecture follows the following design principles:

1. **Microservices-first:** Loosely coupled services that can be developed, deployed, and scaled independently.
2. **Cloud-native:** Leveraging AWS managed services (EKS, RDS, S3, MSK) for reduced operational overhead.
3. **Offline-first:** Local data persistence and sync capabilities for 30-40% rural areas with intermittent connectivity.
4. **IoT-first:** Real-time ingestion and processing of telemetry from 10,000+ cold chain sensors.
5. **Logistics-first:** Route optimization and warehouse management algorithms for perishable produce.
6. **Security-by-design:** Encryption at rest and in transit, RBAC, comprehensive audit logging.
7. **API-first:** Well-documented RESTful APIs for all external integrations (QSRs, kiranas, processors).

1.1 User Personas Supported

The architecture supports the following key user personas:

1.2 High-Level Architecture Diagram

The architecture comprises six layers:

1. **Client Layer:** Mobile apps (farmers, field officers), B2B customer portal (web), Unnati SaaS (collection centers), WhatsApp integration.
2. **API Gateway Layer:** Kong/AWS API Gateway for routing, rate limiting, authentication, and observability.
3. **Microservices Layer:** 12+ services including Loan Origination, Credit Engine, IoT Management, Cold Chain Monitor, B2B Order Management, Warehouse Optimization, Logistics Engine, Outgrow Management, and Analytics.
4. **Data Layer:** PostgreSQL (operational), TimescaleDB (time-series), Redis (caching), Snowflake (warehouse), S3 (data lake).
5. **External Integrations:** Payment gateways (Razorpay, Cashfree), government systems (DBT, Aadhaar), third-party APIs (Google Maps, Twilio, weather).
6. **Deployment & Observability:** EKS on AWS, CI/CD (GitHub Actions + ArgoCD), monitoring (Datadog, Prometheus, ELK).

Table 1: User Personas and Capabilities

Persona	Capabilities
Farmers (individual)	Receive working capital loans (produce-linked), track repayment schedule, access advisory (pest alerts, weather), view real-time prices, enroll in Outgrow contract farming.
B2B Customers (Restaurants, Kiranas, QSRs)	Place orders (daily/weekly), track delivery (real-time), view invoice/payment history, access inventory financing, quality rating dashboard.
Collection Center Operators	Onboard farmers, record daily produce volume/quality, manage payments, access WayCool Lens platform, schedule pickups.
Field Officers	Onboard farmers, conduct due diligence, collect repayments, provide training, install IoT sensors (cold chain).
Warehouse Managers	Inventory management, quality checks, dispatch planning, temperature monitoring (ambient/chilled/frozen).
Logistics Managers	Route optimization, driver tracking, vehicle maintenance, fuel efficiency monitoring.
Corporate Users	Access data APIs, purchase insights (supply-demand forecasts, price trends), monitor portfolios.
WayCool Admins	Underwrite loans, approve farmers, monitor KPIs, manage NBFC compliance, configure system.

2 Client Layer

2.1 Mobile Application (Android & iOS)

Technology: React Native (single codebase for Android/iOS) with TypeScript.

2.2 WhatsApp Integration

Primary channel for farmer advisory (high engagement, low friction).

- Technology: Meta WhatsApp Business API (Cloud API) + Gupshup for regional language NLP.
- Use cases: Pest alerts (upload photo → AI diagnosis → remedy within 30 sec), weather forecasts (7-day), price alerts (mandi vs. WayCool), repayment reminders, loan approval notifications.
- Language support: NLP for Hinglish and 12 regional languages using Meta’s multilingual model.
- Fallback: Twilio SMS when WhatsApp unavailable or user opts out.

Table 2: Mobile App Features by User Type

User Type	Key Features
Farmer App	Loan application & tracking (balance, due date, repayment history), produce price discovery (real-time mandi & WayCool rates), advisory (pest alerts via image recognition, weather forecasts), Outgrow contract farming enrollment, insurance enrollment.
Field Officer App	Farmer onboarding (Aadhaar eKYC, geotagged photos), repayment collection (offline with digital receipt), IoT sensor installation checklists, training module (videos in regional languages).
B2B Customer App	Order placement (daily/weekly recurring), delivery tracking (real-time GPS), invoice download, quality rating (rate each delivery), inventory financing application.

2.3 B2B Customer Portal (Web)

Technology: React.js with TypeScript, Tailwind CSS.

- Features: Dashboard (order history, spending trends, quality scores), order management (daily ordering with cut-off at 8 PM), delivery tracking (real-time GPS), invoice management (download GST invoices), inventory financing application (pre-approved limits), analytics (price comparisons, savings vs. mandi).
- Integrations: ERP systems (Tally, Zoho) via API, POS systems for kirana stores.

2.4 Unnati SaaS (Collection Center Portal)

Technology: React.js + Python Django backend.

- Features: Farmer register (Aadhaar-linked with biometric authentication), daily produce procurement recording (volume, quality grade A/B/C), payment calculation (instant), loan application tracking, GST/TDS compliance (automated filing), IoT sensor dashboard (temperature/humidity).
- User base: 850+ collection centers currently → 5,000+ by FY30F.

2.5 Call Center Integration

Technology: Ozonetel / Exotel (cloud telephony).

- Agent count: 30+ agents by FY27, 100+ by FY30.
- Features: IVR in 12 regional languages, automated call-back for missed calls, agent dashboard with customer 360° view (loan history, order history, repayment status), AI-suggested responses (LLM-powered), call recording and quality assurance.

3 API Gateway Layer

The API Gateway serves as the single entry point for all client requests.

Table 3: API Gateway Configuration

Component	Specification
Technology	Kong / AWS API Gateway
Authentication	JWT+OTP (farmers); API keys (corp); OAuth2 (B2B)
Rate Limiting	100/min (farmer); 1k/min (B2B); 10k/min (int); 5k/min (API)
Routing	/loans, /orders, /inventory, /logistics, /farmers, /data, /iot
SSL/TLS	TLS 1.3 + auto cert renewal
DDoS	AWS Shield + WAF (OWASP Top 10)
Validation	JSON schema; size limits (1MB farmer, 10MB corp)
Logging	ELK stack; logs retained 90 days
Caching	Redis (catalog, prices, routes)

3.1 API Versioning Strategy

- URI versioning: /v1/orders, /v2/orders.
- Deprecation policy: Announce 6 months in advance; maintain backward compatibility for 12 months.
- Current versions: v1 (stable), v2 (beta for new features).

4 Microservices Layer

The microservices layer comprises loosely coupled services that can be scaled independently based on load. Services communicate via gRPC (synchronous) and Kafka (asynchronous events).

4.1 Microservices Summary

Table 4: Microservices Overview

Service	Stack	Prior	Function
Loan Origination	Java SB	H	Farmer loans
Credit Engine (WCE)	Python/ML	H	ML scoring
Order Management	Node.js	H	B2B orders
Warehouse Optimization	Python	M	Inventory mgmt
Logistics Engine	Go	M	Route optimization
IoT Management	Python+MQTT	M	Sensor telemetry
Cold Chain Monitor	Node.js	M	Temp alerts
Outgrow Management	Python Django	M	Contract farming
NBFC Lending	Java SB	H	Collections
Farmer Advisory	Python FastAPI	L	WhatsApp advisory
Data Monetization	Python FastAPI	L	Enterprise APIs
Analytics	Python+dbt	M	BI dashboards

4.2 Service 1: Loan Origination & Underwriting Service

Handles the complete loan lifecycle for farmer working capital financing.

- Technology: Java Spring Boot 3.x, Drools (rules engine for loan structuring).
- Database: PostgreSQL (read-write) + Redis (caching for frequent queries).
- Key functions:
 - Farmer onboarding: Digital application with e-sign, Aadhaar-based KYC (UIDAI API), bank account verification (penny drop).
 - Credit underwriting: ML model using 50+ features (produce sales volume, quality consistency, payment history, cultivation area via satellite).
 - Loan structuring: Tenor (6-12 months), interest rate (12-16%), repayment schedule (daily/weekly produce deduction).
 - Disbursement: Direct to farmer's bank account or as input credit to verified suppliers.
 - Repayment tracking: Automated deduction from produce payment (integration with collection center procurement system).

Table 5: Credit Scoring Model - Technical Specifications

Parameter	Value
Model architecture	XGBoost + LSTM (time-series)
Training data	1.5M+ farmer-months
Number of features	52
AUC-ROC	0.87
Default prediction accuracy (3 months)	78%
Inference latency (p95)	400 ms
Feature freshness	Real-time (daily produce data)
Retraining frequency	Weekly (automated)

4.2.1 WayCool Credit Engine (WCE) - Model Architecture

4.3 Service 2: Order Management Service

Handles B2B order placement, fulfillment, and invoicing for 60,000+ customers.

- Technology: Node.js + PostgreSQL + Redis (caching).
- Key functions:
 - Order placement: Daily/weekly recurring orders, cut-off at 8 PM for next-day delivery.
 - Inventory allocation: Real-time inventory check across 35 warehouses, allocation by FIFO.
 - Pricing engine: Dynamic pricing based on procurement cost, demand, customer tier (volume discounts).
 - Invoicing: Automated GST invoice generation (GSTR-1, GSTR-3B), integration with Tally/Zoho.
 - Delivery tracking: Real-time GPS tracking with ETA (estimated time of arrival) updates every 5 minutes.

4.4 Service 3: Warehouse Optimization Service

Manages inventory across 35+ warehouses (ambient, chilled, frozen).

- Technology: Python + PostgreSQL + Redis (geospatial).
- Key functions:
 - Putaway optimization: Assign incoming produce to optimal bin based on temperature requirement, shelf life, and existing inventory.
 - Picking optimization: Wave picking for B2B orders, route optimization within warehouse (shortest path).
 - Inventory tracking: Real-time by SKU, batch, and location; FIFO/FEFO (first-expiry-first-out) for perishables.

- Replenishment alerts: Auto-alert when inventory falls below safety stock (based on forecasted demand).
- Shelf life monitoring: Track days since harvest; auto-discard when expired.

4.5 Service 4: Logistics Engine

Optimizes route planning and tracks delivery vehicles.

- Technology: Go + Google OR-Tools (vehicle routing problem solver) + PostgreSQL + Redis.
- Key functions:
 - Route optimization: VRP (Vehicle Routing Problem) with time windows, capacity constraints, and delivery priority (hotel breakfast vs. kirana afternoon). Reduces distance by 18% and fuel cost by 12%.
 - Driver assignment: Match drivers to routes based on location, vehicle type (reefer vs. dry), and availability.
 - Real-time tracking: GPS tracking for 1,200+ vehicles, ETA updates every 5 minutes, geofencing for delivery confirmation.
 - Fuel efficiency monitoring: Track km/L, idle time, route deviations.
 - Vehicle maintenance: Predictive maintenance based on engine hours, distance, and fault codes.

4.6 Service 5: IoT Device Management Service

Manages cold chain sensors (temperature/humidity) across warehouses and reefer vehicles.

- Technology: Python + MQTT broker (Eclipse Mosquito) + AWS IoT Core.
- Protocols: MQTT (telemetry), HTTPS (configuration).
- Sensor count: 500+ currently → 10,000+ by FY30F (warehouses + reefer vehicles).
- Key functions:
 - Device provisioning: OTA configuration, firmware updates (signed binaries).
 - Data ingestion: Real-time temperature/humidity (sampled every 30 seconds, batched every 5 minutes).
 - Device health monitoring: Battery level (for battery-operated sensors), connectivity status, last heartbeat.
 - Anomaly detection: ML-based detection of temperature excursions (spoilage risk prediction).
 - Command dispatch: Remote configuration of alert thresholds, reboot.

Table 6: IoT Sensor Telemetry Schema

Field	Description
sensor_id	Unique device identifier
timestamp	UTC time of measurement
temperature_c	Temperature in Celsius (0.1°C resolution)
humidity_percent	Relative humidity (1% resolution)
location_lat	Latitude (GPS)
location_lng	Longitude (GPS)
vehicle_id	Vehicle identifier (if mobile)
warehouse_id	Warehouse identifier (if static)
battery_voltage	Device battery level (V)
rsi	Cellular signal strength (dBm)

4.7 Service 6: Cold Chain Monitoring Service

Monitors temperature and humidity across the cold chain to prevent spoilage.

- Technology: Node.js + MQTT + InfluxDB (time-series database).
- Key functions:
 - Real-time temperature monitoring: Warehouses (4°C target for chilled, -18°C for frozen), reefer vehicles (temperature every 5 minutes).
 - Alert generation: SMS/WhatsApp alert when temperature exceeds threshold (e.g., >6°C for 30 minutes) — average response time <10 seconds.
 - Spoilage prediction: ML model predicts produce spoilage risk based on temperature time-series (AUC 0.85).
 - Compliance reporting: Automated FSSAI compliance reports for B2B customers (daily temperature logs).
 - Energy efficiency: Compressor duty cycle analysis to identify inefficient reefer units.

4.8 Service 7: Outgrow Contract Farming Service

Manages WayCool’s contract farming program with pre-agreed prices and input financing.

- Technology: Python Django + Celery (async tasks) + PostgreSQL.
- Farmer coverage: 120,000+ currently → 2 million by FY30F.
- Key functions:
 - Contract enrollment: Farmer signs contract (digital signature) agreeing to pre-agreed price (MSP + 5-10% premium).
 - Input financing: Seeds, fertilizer, pesticides provided upfront; loan automatically created in Loan Origination Service.
 - Crop advisory: Weekly agronomist visit scheduling, pest alerts via WhatsApp.

- Harvest planning: Predict harvest date based on sowing date and weather; coordinate logistics for pickup.
- Price settlement: Automatically calculate payment (volume $\times$ agreed price), deduct loan repayment, disburse balance to farmer.

4.9 Service 8: B2B Customer Portal Service

Powers the web portal for restaurants, kiranas, and QSRs.

- Technology: Python FastAPI + PostgreSQL + Redis.
- Customer base: 25,000+ currently $\rightarrow$ 60,000+ by FY30F.
- Key functions:
 - Catalog management: Real-time inventory visibility (quantity, price, quality grade), dynamic pricing.
 - Order management: Daily/weekly recurring orders, order history, invoice download.
 - Delivery tracking: Real-time GPS tracking with ETA, delivery confirmation with photo.
 - Quality feedback: Rate each delivery (1-5 stars); ratings used for farmer scoring and logistics performance.
 - Inventory financing: Pre-approved working capital loans (Rs.50,000-2,00,000) based on order history and payment track record.

4.10 Service 9: Farmer Advisory Service

Delivers personalized advisory to farmers via WhatsApp and SMS.

- Technology: Python FastAPI + AI/ML (GPT-based LLM for localized content generation).
- Advisory volume: 5M+ messages/month by FY30F.
- Key functions:
 - Pest alerts: Image recognition (upload photo $\rightarrow$ disease/pest identification $\rightarrow$ remedy within 30 seconds). Accuracy 85% for common pests.
 - Weather forecasts: 7-day forecast with hyperlocal (village-level) recommendations.
 - Crop planning: Recommended sowing windows based on weather and market prices.
 - Input marketplace: Seeds, fertilizer, pesticides delivered within 48 hours (8-12% commission).
 - Price alerts: Real-time mandi prices vs. WayCool offered price.

4.11 Service 10: Data Monetization Service

Sells anonymized, aggregated data to enterprise customers via API.

- Technology: Python FastAPI + Apache Spark (data aggregation) + Snowflake.
- Data volume: 5 TB raw data per month → 50 TB by FY30F.
- Key functions:
 - Data anonymization: Remove PII (name, Aadhaar, mobile); aggregate at district/block level (minimum 100 farmers per cohort).
 - API access: RESTful APIs with API keys; rate limiting by subscription tier.
 - Report generation: Scheduled reports for government (Ministry of Food Processing) and enterprise customers.
 - Datasets available: Supply-demand forecasts by region/crop, price trends (historical + predictive), quality metrics by region, logistics efficiency benchmarks.

4.12 Service 11: Analytics & Reporting Service

Provides business intelligence and operational dashboards.

- Technology: Apache Superset / Power BI + dbt (data transformation) + Snowflake.
- Dashboard refresh: Real-time (operational) / Daily (analytical).
- Key dashboards:
 - Executive dashboard: Revenue, GMV, gross margin, ROA, ROE, farmer growth, B2B customer growth.
 - Operations dashboard: Order fulfillment rate, on-time delivery percentage, spoilage percentage by warehouse.
 - Credit dashboard: Loan application volume, approval rate, AUM, GNPA trends.
 - Logistics dashboard: Fleet utilization (
 - Cold chain dashboard: Temperature compliance by warehouse/vehicle, spoilage alerts by route.
 - Impact dashboard: Farmers served (cumulative), income uplift estimated, women farmers percentage.

4.13 Service 12: Notification Service (Cross-cutting)

Handles all outbound communications across channels.

- Technology: Node.js + Bull (queue) + channel adapters.
- Channels: WhatsApp (primary), SMS (fallback), Email (corporate), In-app (push).
- Queue management: Priority queues (high: OTP, order confirmation, repayment reminders; medium: delivery updates; low: marketing).

- Delivery tracking: Delivery status, open rates, click-through rates.
- Templates: 50+ localized templates in 12 languages.

5 Data Layer

The data layer consists of multiple storage systems optimized for different use cases (polyglot persistence).

5.1 Operational Databases

Table 7: Operational Databases

Database	Technology	Purpose	Size (FY30F)
Primary OLTP	PostgreSQL 15	Farmers, loans, orders, B2B	15 TB
Time-series	TimescaleDB	IoT telemetry (temp, humidity)	30 TB
Geospatial	PostGIS	Warehouses, delivery routes	500 GB
Caching	Redis (cluster)	Sessions, rate limits, catalog	150 GB
Search	Elasticsearch	Farmer/customer search, logs	3 TB

5.2 Data Warehouse

- Technology: Snowflake (primary) / Amazon Redshift (legacy).
- Data sources: All microservices stream data to Kafka (40+ topics) → transformed via dbt (80+ models) → loaded to warehouse.
- Data volume: 5 TB raw data per month currently → 50 TB by FY30F.
- Refresh frequency: Real-time (streaming) for operational dashboards; daily for analytical reports (3 AM IST).
- Schema: Star schema with fact tables (order facts, loan facts, delivery facts) and dimension tables (farmer dim, customer dim, date dim, product dim).

5.3 Data Lake

- Technology: Amazon S3 (with Glacier Deep Archive for cold storage).
- Content:
 - Raw IoT telemetry (temperature, humidity) — retained 30 days in S3 Standard, 7 years in Glacier Deep Archive.
 - Farmer KYC documents (Aadhaar, land records) — encrypted at rest (AES-256), retained 7 years post-last-transaction.
 - Order and delivery logs — retained 7 years (RBI requirement for financing).
 - System logs (application, access, audit) — retained 30 days in hot, 90 days in warm.
 - ML model artifacts — versioned, retained indefinitely.

5.4 Data Pipeline Architecture

- Ingestion: Kafka (40+ topics) for real-time streaming from all microservices.
- Processing: Apache Spark for batch (nightly aggregations) + Kafka Streams for stream (real-time alerts).
- Transformation: dbt (80+ models) for data warehouse transformations (SQL-based).
- Orchestration: Apache Airflow for scheduled jobs (DAGs for ETL, report generation, ML retraining).
- Quality: Great Expectations for data validation (automated testing of data quality rules).

5.5 Data Governance & Security

- Encryption: AES-256 at rest (S3, RDS, Snowflake); TLS 1.3 in transit (all services).
- Access control: Row-level security in PostgreSQL; column-level security in Snowflake; IAM roles for AWS services.
- Data classification: Public (price trends), Internal (operational metrics), Confidential (farmer PII), Restricted (loan details, Aadhaar).
- Compliance: RBI NBFC regulations (audit trail, data retention), DPDP Act (India), FSSAI (quality data), ISO 27001 (planned FY28).
- Backup: Automated daily snapshots (retained 30 days); cross-region replication to ap-southeast-1 (Singapore) for DR.

6 External Integrations

6.1 Payment Gateways

Table 8: Payment Gateway Integrations

Gateway	Use Case	Volume (FY30F)
Razorpay	B2B payments (UPI, cards, net banking)	Rs.1,000 Cr/mo
Cashfree	Farmer payouts, field officer commissions	Rs.200 Cr/mo
Bank API (ICICI/HDFC/SBI)	Loan disbursement, NACH collection	Rs.300 Cr/mo

6.2 Government Systems

Table 9: Government Integrations

System	Purpose
Aadhaar eKYC (UIDAI)	Farmer identity, age/address, biometric
DigiLocker	Land records, farmer certificate
GST portal (NIC)	Invoice validation, GSTR filing
IMD (Met dept)	Weather (7-day), pest alerts, rainfall
DBT (NPCI)	Subsidies (PM-KISAN, fertilizer)
Ministry of Food Processing	Scheme reporting, cold chain claims

6.3 Third-Party Services

- Google Maps API: Delivery route optimization, geocoding, distance matrix for logistics planning.
- AWS IoT Core: IoT sensor connectivity (MQTT), message brokering, device shadow.
- Twilio: SMS notifications (fallback when WhatsApp unavailable; 2M+ messages/month).
- Sentry: Error tracking and performance monitoring.
- Datadog: Infrastructure monitoring (CPU, memory, network) and alerting.
- Cloudflare: CDN for static assets (web portal, mobile app downloads) and DDoS mitigation.
- WhatsApp Business API (Meta/Gupshup): Farmer advisory, order notifications, repayment reminders.

7 Scalability & Performance

7.1 Scaling Strategy

The architecture is designed to scale horizontally to handle 5x growth by FY30F.

Table 10: Horizontal Scaling Targets by FY30F

Component	FY26F	FY30F	Growth
Active farmers (million)	0.85	4.0	4.7x
Active B2B customers	25,000	60,000	2.4x
IoT sensors deployed	500	10,000	20x
Daily IoT data points	50K	1M	20x
Daily API requests	500,000	5,000,000	10x
Daily orders	25,000	100,000	4x
Data warehouse size (TB)	5	50	10x

7.2 Auto-scaling Configuration

- Orchestration: EKS (Amazon Elastic Kubernetes Service) with Horizontal Pod Autoscaler (HPA) and Cluster Autoscaler.
- Trigger conditions: CPU >70%, memory >80%, custom metric (requests per second >1,000, Kafka consumer lag >10,000).
- Cooldown period: 3 minutes (scale up), 10 minutes (scale down).
- Node groups:
 - Compute-optimized (c5.large/c5.xlarge): Loan origination, credit engine, order management.
 - Memory-optimized (r5.large/r5.xlarge): Redis caching, PostgreSQL, Elastic-search.
 - GPU-optimized (g4dn.xlarge): ML model inference (credit scoring, spoilage prediction).
 - IoT-optimized (m5.large): MQTT broker, telemetry ingestion.

7.3 Performance SLAs

Table 11: Performance Service Level Agreements (SLAs)

Service	Target Response Time (p95)	Current (p95)
Order placement	<1 sec	0.7 sec
Credit decision (automated)	<5 sec	3.0 sec
IoT data ingestion	<500 ms	280 ms
Warehouse dashboard load	<2 sec	1.3 sec
API data access (aggregated)	<1 sec	0.6 sec
Cold chain alert generation	<10 sec	4 sec
Mobile app sync	<3 sec	1.8 sec
Route optimization	<30 sec	15 sec

8 Deployment Architecture

8.1 AWS Infrastructure

Table 12: AWS Infrastructure Components

Component	Configuration
Compute (K8s)	EKS — 20-100 nodes (c5.large, r5.large, g4dn.xlarge, m5.large)
Database	RDS PostgreSQL Multi-AZ, db.r5.2xlarge (16 vCPU, 64 GB RAM)
Time-series	TimescaleDB managed on RDS, 10 TB storage
Caching	ElastiCache Redis cluster (3 shards, 2 replicas each)
Object storage	S3 — 50 TB standard, 200 TB Glacier
Data warehouse	Snowflake X-Small to 2X-Large (auto-scaling)
Streaming	MSK (Kafka) — 6 brokers (kafka.m5.2xlarge)
CDN	CloudFront — 15+ edge locations in India
Networking	VPC CIDR 10.0.0.0/16, 6 subnets (3 public, 3 private)

8.2 Primary Region: Mumbai (ap-south-1)

- Production workloads: All microservices, databases, data warehouse.
- Availability Zones: 3 AZs (ap-south-1a, ap-south-1b, ap-south-1c) for high availability.
- Capacity: 100 nodes (peak), 50 nodes (average).

8.3 Disaster Recovery

Table 13: Disaster Recovery Objectives

Metric	Target
RTO	4 hrs (critical: 1 hr)
RPO	15 min (critical data: 5 min)
Backup frequency	Daily snapshots (RDS, EBS); continuous S3 replication
DR testing	Quarterly (simulated failover)

- DR Region: Singapore (ap-southeast-1).
- Cross-region replication for critical data (loan book, farmer profiles, order history).
- RDS read replica (async replication).
- S3 cross-region replication (CRR) for KYC documents and transaction logs.
- Warm standby: EKS cluster with minimum nodes (20% capacity).

8.4 CI/CD Pipeline

- Source control: GitHub (monorepo with 15+ microservices).
- CI: GitHub Actions (build, test, lint, security scan).
- CD: ArgoCD (GitOps, automated deployment to EKS).
- Environments: dev → staging → production (canary deployment: 5% → 20% → 100%).
- Rollback: Automated (health check failure) or manual (one-click via ArgoCD).
- Build frequency: 40+ builds per day.

9 Offline-First Design

Given that 30-40% of farmers and collection centers operate in areas with 2G/3G or intermittent 4G connectivity, WayCool implements offline-first design principles.

9.1 Offline Capabilities by Component

Table 14: Offline Functionality Matrix

Component	Offline	Max Duration	Sync Strategy
Farmer onboarding	Yes	7 days	Batch sync (delta)
Loan application (draft)	Yes	30 days	Submit online (idempotent)
Produce volume recording	Yes	24 hrs	Priority queue upload
Repayment (field officer)	Yes	24 hrs	Upload with receipt
Order placement (B2B)	Partial	12 hrs	12h cache, stale-while-revalidate
IoT telemetry (sensors)	Yes	30 days	Batch upload (MQTT)
Delivery proof (photo)	Yes	7 days	Upload on connectivity

9.2 Local Storage Architecture

- Mobile app (farmer/field officer): SQLite (up to 500 MB) for farmer data, loan applications, offline forms.
- IoT sensors: Flash storage (32 MB) for telemetry; store-and-forward for up to 30 days (20,000 readings).
- Collection center tablets: SQLite + file storage for produce records (photos, weight logs).

9.3 Sync Conflict Resolution

Table 15: Sync Conflict Resolution

Scenario	Resolution
Profile preferences	Last-write-wins (timestamp)
Loan application	Server wins; idempotency key prevents duplicates
Produce volume	Collection center wins (trusted); farmer dispute flag
Repayment collection	Server wins; field officer reconciles offline
Order placement	Server wins; idempotency key blocks duplicates
Delivery proof	Server wins (customer confirm); driver retries with backoff

10 Security Architecture

10.1 Authentication & Authorization

Table 16: Authentication Methods by User Type

User Type	Method	Controls
Farmers	OTP (mobile + Aadhaar)	5 min expiry; max 3 attempts
B2B Customers	JWT + password	2FA optional; 8 hr timeout
Collection Ctr Ops	JWT + geo-attendance	Within 500m of center
Field Officers	JWT + password	12 hr timeout
Corporate Users	API keys + IP whitelist	Key rotation 90 days
Admins	MFA (TOTP) mandatory	2 hr timeout; audit logs
Internal Services	mTLS (service mesh)	SPIFFE; auto cert rotation

10.2 RBAC Roles and Permissions

Table 17: RBAC Roles (7 roles)

Role	Permissions
Farmer	View own loans, repayments, advisory, orders
B2B Customer	View orders, delivery tracking, invoices, apply for inventory financing
Collection Center	Manage farmers at assigned center, record produce, view loans
Field Officer	Farmer onboarding, repayment collection, IoT installation
Credit Officer	Loan underwriting, credit decision, limit setting
Admin	User management, system configuration, report access
Super Admin	Full system access, audit log review, role assignment

10.3 Security Controls Summary

10.4 IoT Device Security (Cold Chain Sensors)

- Device identity: X.509 certificates (per device) issued by AWS IoT Core.
- Mutual TLS (mTLS): Device authenticates to cloud; cloud authenticates to device.
- Firmware signing: OTA updates only with signed binaries; rollback protection.
- Tamper detection: Physical tamper switch (triggered when enclosure opened).
- Data validation: Anomaly detection for impossible values (temperature $>50^{\circ}\text{C}$ in chilled warehouse).

10.5 Compliance Mapping

Table 18: Security Controls

Control	Implementation
Authentication	OTP (farmers); JWT + MFA (admins); mTLS (services)
Authorization	RBAC (7 roles); ABAC for fine-grained (region, warehouse)
API Security	JWT validation at gateway; rate limiting by role; input validation
Data Encryption (Rest)	AES-256 (RDS, S3, Snowflake, Elasticsearch)
Data Encryption (Transit)	TLS 1.3 (all services); mTLS for service-to-service
Secrets Management	AWS Secrets Manager (API keys, DB credentials, certs)
Vulnerability Scanning	Weekly automated (AWS Inspector, Snyk)
Penetration Testing	Annual third-party (CERT-In empaneled firm)
Audit Logging	All API calls logged (user ID, timestamp, IP, endpoint)
IAM	Least privilege principle; temporary credentials (STS)
WAF	AWS WAF (OWASP Top 10, IP reputation, SQL injection)

11 Monitoring & Observability

11.1 Monitoring Stack

11.2 Key Metrics to Monitor

11.2.1 Business KPIs (Dashboard - 5 min refresh)

- Revenue (daily, weekly, monthly by region), GMV, gross margin.
- Order fulfillment rate (%), on-time delivery percentage.
- Loan disbursement volume (daily), GNPA trends, repayment rate.
- Farmer onboarding rate (daily) and retention (72% target).
- B2B customer acquisition and churn.
- IoT sensor health (% online, battery <20%).
- Spoilage percentage (by warehouse, by route).

11.2.2 Technical KPIs (Dashboard - real-time)

- API p95 latency by endpoint (target: <1 sec).
- Error rate (HTTP 5xx) — target <0.1% (99.9% availability).
- Credit engine inference latency — target <400 ms (p95).
- IoT data ingestion lag — target <1 second (device to database).
- Database connection pool utilization (target <80%).
- Kafka consumer lag — target <10,000 messages.
- Pod restart count (by service).
- Node CPU/memory utilization (target <70%).

Table 19: Compliance Mapping (RBI, DPDP, FSSAI)

Requirement FSSAI	RBI	DPDP
Data localization (India only)		—
Consent management	—	
Data retention (7 years)		—
Right to be forgotten	—	
Data breach notification (72 hours)	—	
Audit trail (all transactions)		—
Quality reporting	—	—
Encryption (AES-256, TLS 1.3)		

Table 20: Monitoring Tools

Tool	Purpose
Datadog	Infrastructure metrics (CPU, memory, network)
Prometheus + Grafana	App metrics (latency, errors, business KPIs)
ELK Stack	Centralized logging (app, audit, access logs)
AWS X-Ray	Distributed tracing for microservices
Sentry	Error tracking, crash reporting
PagerDuty	On-call alerting, incident management
AWS CloudTrail	AWS API audit logging
AWS Config	Resource compliance monitoring

11.3 Alerting Rules

- Critical (P0): API error rate >1% for 5 min → PagerDuty to on-call engineer.
- High (P1): Cold chain temperature >6°C for 30 min → SMS to warehouse manager.
- Medium (P2): Database connection pool >80% for 10 min → Slack to engineering team.
- Low (P3): IoT sensor offline for >24 hours → email to field ops.

12 Cost Optimization

12.1 Current & Projected Cloud Spend

Table 21: Cloud Spend Projection (Rs. Cr)

Category	FY26F	FY28F	FY30F
Compute (EKS, EC2, Lambda)	3.5	7.0	14.0
Storage (RDS, S3, Snowflake, EBS)	1.5	3.0	6.0
IoT connectivity (AWS IoT Core)	0.5	1.5	3.0
Data transfer (in/out, CloudFront)	0.5	1.0	2.0
Third-party APIs (maps, weather, SMS)	1.0	2.0	4.0
Total Cloud Spend	7.0	14.5	29.0

Note: Cloud spend as % of revenue declines from 0.25% (FY26F) to 0.24% (FY30F) — significantly lower than Stellapps because WayCool’s revenue is product sales (higher denominator).

12.2 Optimization Strategies

1. **Reserved Instances (RI):** Purchase 3-year RIs for stable workloads (control plane, databases, Redis) → 35% savings.
2. **Spot instances:** Use for non-production (staging, dev) and batch processing (data aggregation, ML training) → 65% savings.
3. **S3 lifecycle policies:** Transition IoT data to Glacier Instant Retrieval after 30 days → 80% storage cost reduction.
4. **CloudFront caching:** Cache web portal assets and mobile app downloads at edge → 70% reduction in origin fetches.
5. **Data compression:** Compress IoT telemetry (gzip, 70% compression) → 50% reduction in data transfer costs.
6. **Serverless for low-traffic:** Use Lambda for infrequent operations (report generation, data exports) → pay-per-execution.

13 Conclusion

The WayCool system architecture is designed to support the company's ambitious growth targets from Rs.2,800 crore (FY26E) to Rs.12,000+ crore (FY30F) while maintaining high availability (99.9% uptime target), sub-second response times for critical operations (order placement, credit decision), and full compliance with RBI NBFC regulations, DPDP Act (India), and FSSAI quality standards.

13.1 Key Architectural Decisions Summary

Table 22: Key Architectural Decisions & Rationale

Decision	Rationale
Microservices over monolithic	Independent scaling (order mgmt vs. credit engine)
EKS (K8s) over ECS	Portability, HPA, larger community
PostgreSQL + TimescaleDB	Single DB for transactional + time-series
MQTT for IoT	Lightweight, offline support, QoS
Offline-first mobile	30-40% rural areas with intermittent connectivity
Snowflake for analytics	Compute/storage separation, auto-scaling
Kafka for streaming	Decouple services, replay, 40+ topics
React Native	Single codebase (Android/iOS)
Go for logistics	Performance, concurrency, compiled

13.2 Projected Capacity by FY30F

- Farmers: 4 million+ (4.7x growth from 0.85M).
- B2B Customers: 60,000+ (2.4x growth from 25,000).
- Collection Centers: 5,000+ (5.9x growth from 850).
- Warehouses: 50+ (1.4x growth from 35).
- IoT sensors: 10,000+ (20x growth from 500).
- Daily IoT data points: 1 million (20x growth).
- Daily API requests: 5 million (10x growth).
- Farmer financing AUM: Rs.6,000 crore (7x growth).
- Annual revenue: Rs.12,000+ crore (4.3x growth).

The architecture is projected to handle these targets with:

- Uptime: 99.9% (8.76 hours downtime/year).
- P95 latency: <1 sec for critical APIs.
- RTO: 4 hours (DR).
- RPO: 15 minutes (5 min for critical data).

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